

August 31, 2026

Clarius Mobile Health

Dear Hiring Team,

I am writing to apply for the Machine Learning Engineer position at Clarius Mobile Health. I recently completed my M.Sc. in Computer Science at the University of Calgary, where my work focused on machine learning, computer vision, data pipelines, and scalable software for large geospatial and scientific datasets. I am especially interested in Clarius because the role connects ML research with a practical medical imaging product that can make high-quality care more accessible.

My graduate research involved building Python and C++ workflows for data collection, preprocessing, validation, model training, and performance evaluation. One major project focused on digital elevation model super-resolution, where I trained deep ResNet models in PyTorch to reconstruct higher-resolution terrain data from lower-resolution inputs. I built the supporting data pipeline with Python, NumPy, pandas, Rasterio, and Google Earth Engine, and evaluated the models with RMSE, slope, TRI, TPI, and wavelet-based metrics. This work gave me hands-on experience with the full ML loop: preparing data, training models, checking whether evaluation methods were reliable, and improving the workflow when results were too optimistic or brittle.

My second major research area was a DGGS-based framework for managing and retrieving large volumes of spatiotemporal satellite data. I designed multiresolution storage and retrieval methods, implemented performance-sensitive C++ components, and evaluated the system on large Canada-wide datasets. That project strengthened my ability to think about scalability, robustness, and maintainability in data-heavy systems, which I see as directly relevant to improving ML tools, data pipelines, retraining workflows, and production-facing model infrastructure.

I also bring backend engineering experience from Asr Gooyesh Pardaz, where I developed Django REST APIs backed by PostgreSQL for NLP, text-to-speech, and speech-to-text products. I worked on authentication features, payment-service integration, and relational records for service plans and transactions. Alongside my research and teaching assistant work, this helped me become comfortable writing clean code, debugging across systems, documenting technical decisions, and explaining ML and software concepts to technical audiences.

The Clarius role stands out to me because it asks for the kind of applied ML engineering I enjoy most: building models and pipelines carefully enough that they can move from experimentation toward practical use. My background is strongest in Python, PyTorch, computer vision, Linux-based development, scientific data workflows, and research communication. I also have working familiarity with TensorFlow, Docker, AWS, C++, and JavaScript, and I am motivated to keep deepening my production ML deployment and optimization skills.

Thank you for considering my application. I would welcome the opportunity to discuss how my ML, computer-vision, data-pipeline, and software-engineering background could contribute to Clarius's work on robust and accessible ultrasound technology.

Sincerely,

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